

SMARTPARKX

Smart Parking for Smarter Cities

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PROBLEM STATEMENT

SmartParkX

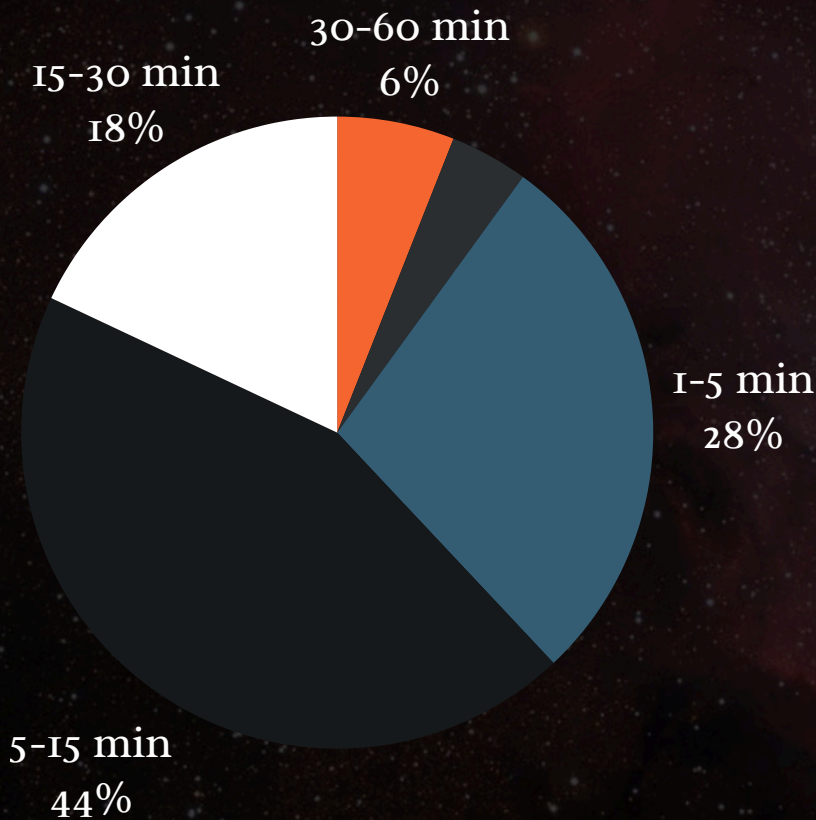
SmartParkX transforms urban parking by combining AI-driven prediction with seamless booking and payment.

It reduces search time, traffic congestion, and fuel waste through a smart, data-driven approach.

WHY?

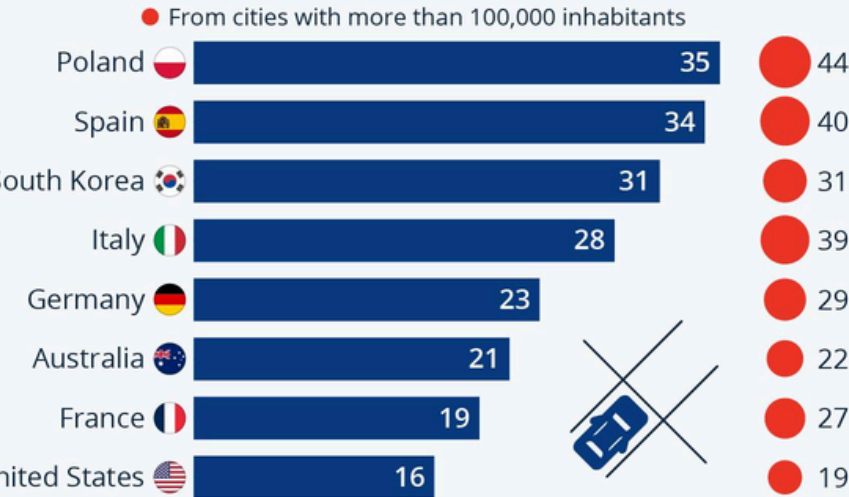
- **Wastes Time:** Drivers spend too long searching for parking
- **Causes Traffic:** Increases congestion and pollution
- **No Smart System:** Lack of prediction and real-time management

Time Spent By Drivers Looking for Parking Per Day



Where Parking Becomes a Headache

Share of respondents saying there were not enough parking spaces where they lived (in percent)



2,000-10,000 respondents (18-64 y/o) surveyed Jan.-Dec. 2023
Source: Statista Consumer Insights



ABSTRACT

WHY?

- Urban areas face severe parking shortages.
- Drivers waste up to 30% of travel time searching for slots.
- Leads to traffic congestion, fuel waste & frustration.

WHAT?

- SmartParkX is a real-time parking prediction and booking platform
- Users can pre-book nearby slots and pay through the app.
- Unified platform combining prediction, booking, and payment.

HOW?

- Azure ML for slot prediction
- Azure Maps for live tracking
- GitHub Copilot for AI-assisted development
- Power BI for analytics dashboard
- GitHub for project management

Impact?

- Reduced congestion & emissions
- Less waiting time for drivers
- Efficient revenue & utilization for operators
- Data-driven insights for city planning

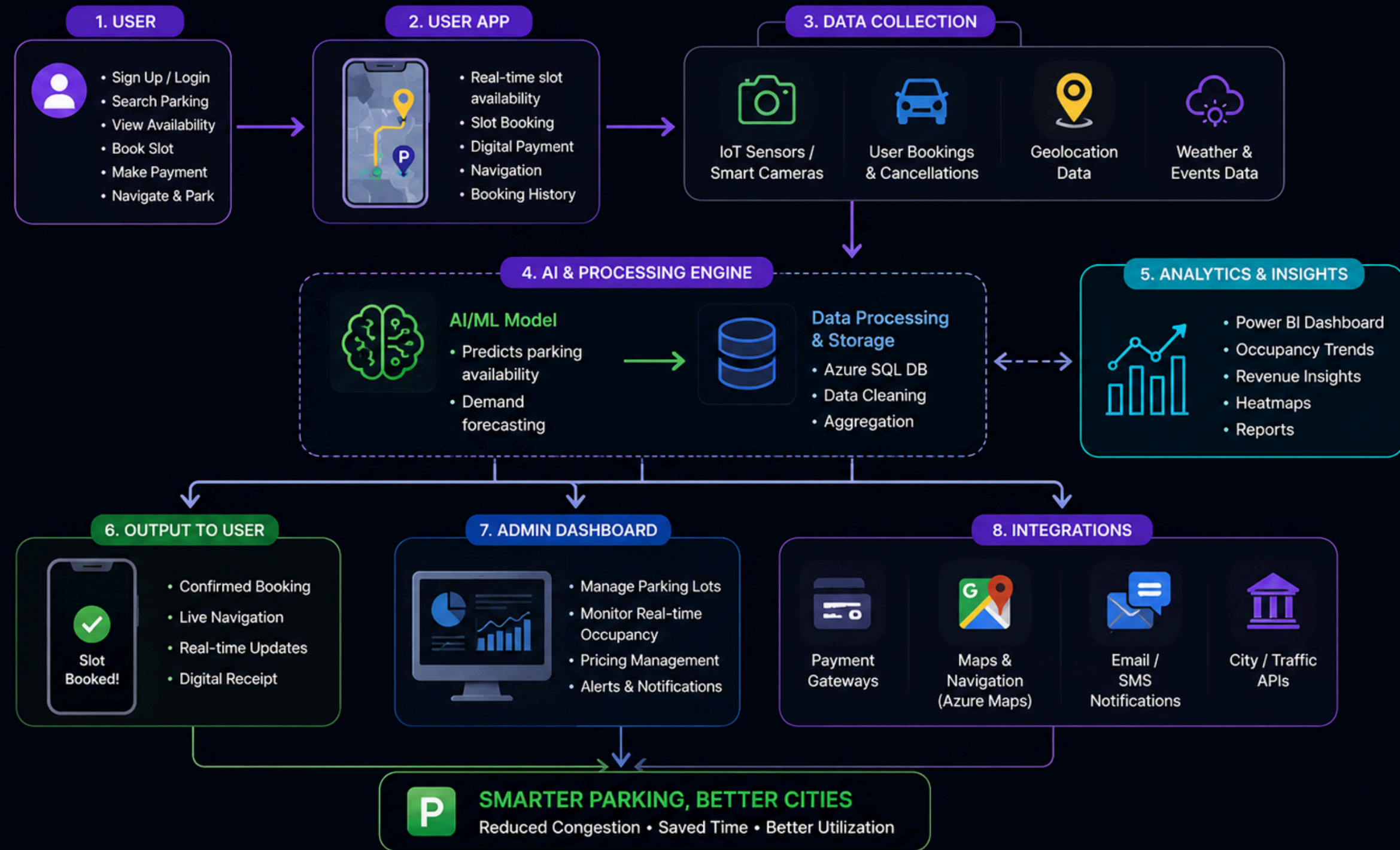


SmartParkX transforms urban parking from a daily frustration into an intelligent, sustainable solution that saves time, reduces emissions, and creates smarter cities for everyone.

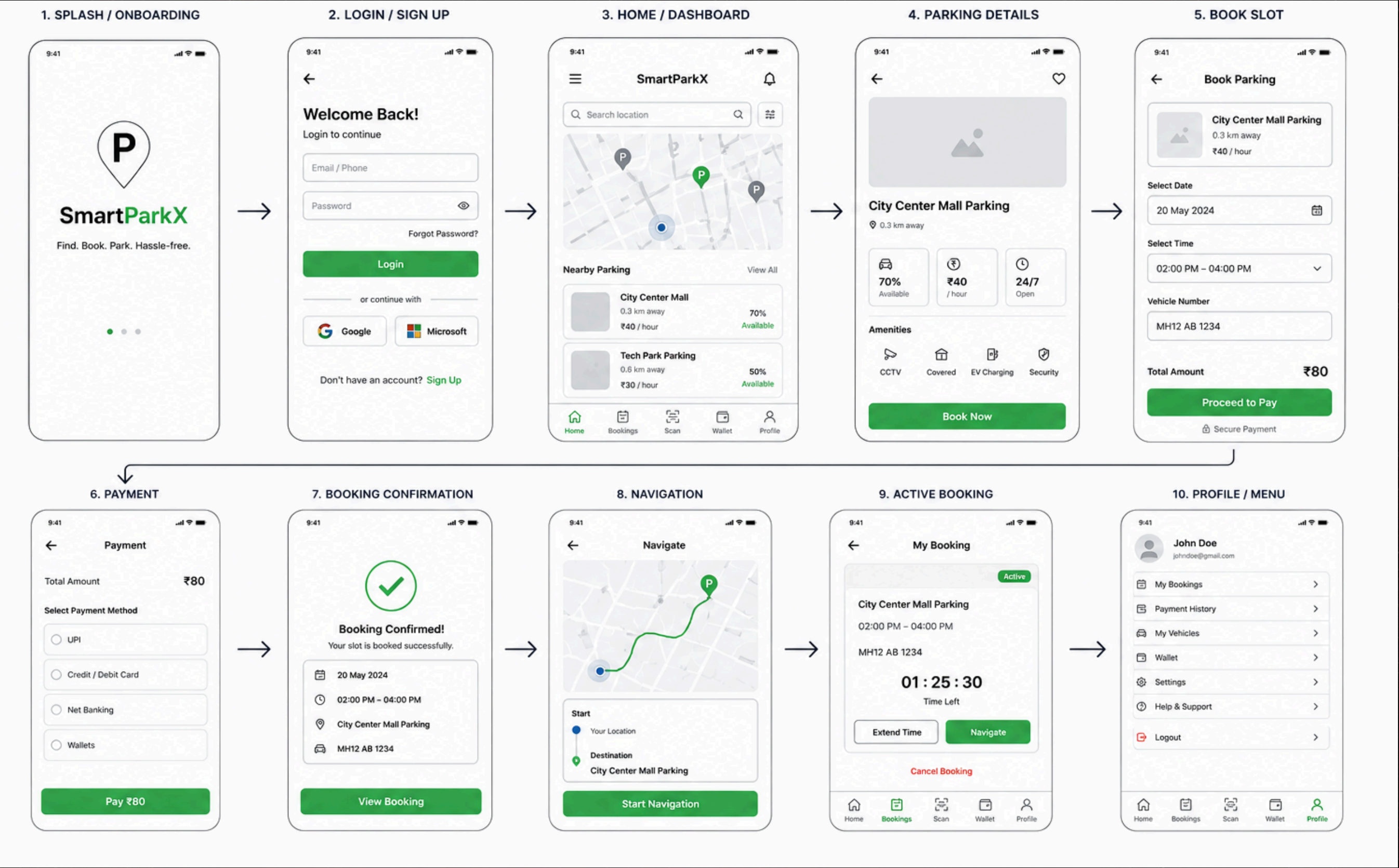
SOLUTION ABSTRACT

- Predicts parking slot availability using AI
- Shows nearby parking spaces in real time
- Allows users to pre-book parking slots
- Uses geofencing for fair and smart booking
- Auto-releases unused slots automatically
- Provides secure digital payment and analytics

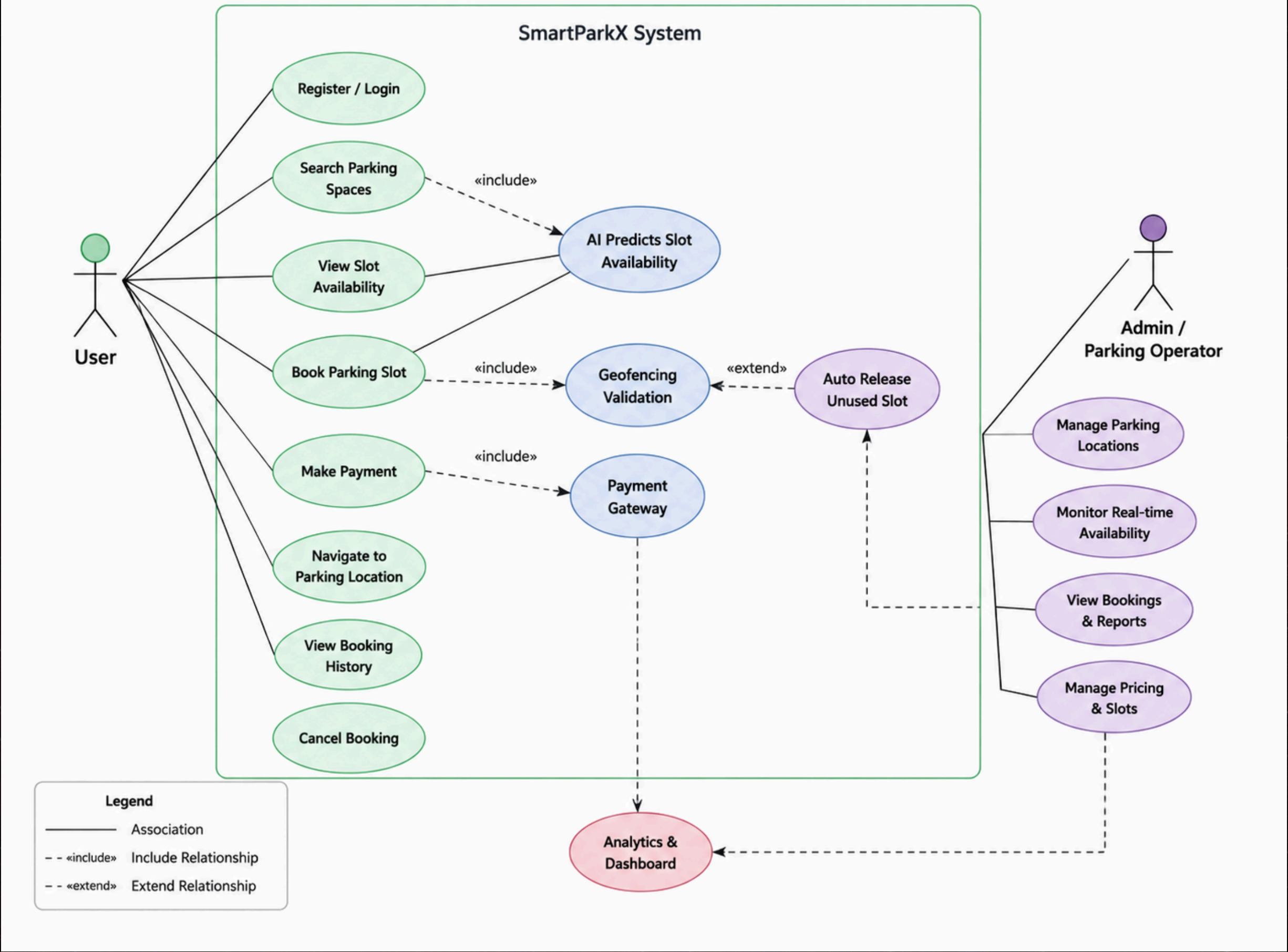
SmartParkX – System Flow



WIREFRAME DIAGRAM



USECASE DIAGRAM



UNIQUE SELLING POINTS

AI-Based Prediction



Predicts parking slot availability in real time using Azure ML.

Smart Geofencing



Allows booking only for nearby users to avoid misuse.



USP



Auto Slot Release

Automatically frees unused slots after timeout.



Integrated System

Combines prediction, booking, navigation, and payment in one platform.

FEASIBILITY

Easy Deployment

- Cloud-based and scalable
- Can be used in malls, offices, and smart cities
- Minimal hardware requirement

Technical Feasibility

- Uses Azure ML and GitHub workflows
- Real-time database and analytics support
- Easily integrable with existing parking systems

Operational Feasibility

- User-friendly mobile interface
- Reduces manual parking management
- Improves efficiency and revenue generation

Conclusion

- Drivers save time and reduce stress.
- Cities see lower congestion and carbon emissions.
- Operators maximize revenue through data-driven insights.

"The future of urban living isn't just about moving faster
—it's about parking smarter."